

Day 318 – Computer Vision Theory

Topic: Images, Pixels, RGB, and Convolution Concept

This day is theory-focused. There is **no coding**. Understanding these concepts will make Days **319–330** (Image Classification, CNNs, Transfer Learning, Object Detection, etc.) much easier.

1. What is Computer Vision?

Computer Vision is a field of AI that enables computers to understand and interpret images and videos, similar to how humans use their eyes and brain.

Examples:

- Face Recognition
 - Self-driving Cars
 - Medical Image Analysis
 - OCR (Text Recognition)
 - Object Detection
 - Image Classification
 - Fingerprint Recognition
-

2. Digital Image

A digital image is simply a matrix of numbers.

Example:

Image



Computer sees

```
[  
  [125,130,140],  
  [135,150,160],  
  [120,140,170]  
]
```

Every number represents a **pixel value**.

3. What is a Pixel?

Pixel = Picture Element

It is the smallest unit of an image.

Example:

5 × 5 Image

□ □ □ □ □

□ □ □ □ □

□ □ □ □ □

□ □ □ □ □

□ □ □ □ □

Total Pixels

25

Suppose an image is

1920 × 1080

Total Pixels

1920 × 1080

=

2,073,600 pixels

Around **2 Million Pixels**

4. Image Resolution

Resolution means

Width × Height

Examples

Resolution Pixels

64×64 4096

128×128 16384

224×224 50176

512×512 262144

Higher resolution means

- More details
- More memory

5. Types of Images

Binary Image

Only two colors

0

or

1

Black and White

Grayscale Image

Only one channel

Values

0

↓

255

0 = Black

255 = White

128 = Gray

Shape

Height × Width

Example

28 × 28

RGB Image

Most common.

Contains

- Red
- Green
- Blue

Each pixel stores

(R,G,B)

Example

(255,0,0)

=

Red

(0,255,0)

=

Green

(0,0,255)

=

Blue

(255,255,255)

=

White

(0,0,0)

=

Black

Shape

Height × Width × 3

Example

224 × 224 × 3

6. RGB Channels

Think of an RGB image as three separate grayscale images stacked together.

Original Image

↓

Red Channel

↓

Green Channel

↓

Blue Channel

Each channel stores intensity from

0

↓

255

7. Image Matrix

Example

3 × 3 Grayscale Image

[

[12 44 89]

[67 90 12]

[34 78 255]

]

Every value represents brightness.

RGB Image

Each cell becomes

(R,G,B)

Example

[(255,0,0)

(100,50,20)

(0,0,255)]

8. Image Size in Deep Learning

Suppose

224×224 RGB

Tensor shape

224

×

224

×

3

Total values

$224 \times 224 \times 3$

=

150,528

One image contains **150,528 numbers**.

9. What is a Tensor?

A Tensor is simply a multi-dimensional array.

Examples

Scalar

7

Vector

[1,2,3]

Matrix

[

[1,2]

[3,4]

]

Image

$224 \times 224 \times 3$

Batch of Images

$32 \times 224 \times 224 \times 3$

10. Why CNN instead of ANN?

Suppose we flatten

$224 \times 224 \times 3$

Input becomes

150,528 neurons

Connecting to just

100 neurons

requires

15 Million+

weights

This is computationally expensive and ignores spatial relationships.

CNN solves this using **convolution**, drastically reducing parameters while preserving image structure.

11. What is Convolution?

Convolution is the process of sliding a small matrix called a **filter (kernel)** over an image to extract useful features.

Instead of looking at the entire image at once, CNN looks at small regions.

Example

Image

↓

Filter slides

↓

Edges

↓

Corners

↓

Textures

↓

Shapes

↓

Objects

12. Kernel (Filter)

Example

3×3 Filter

[
1 0 -1
1 0 -1
1 0 -1
]

Kernel size

3×3

or

5×5

or

7×7

CNN learns these kernels automatically during training.

13. Convolution Example

Image

10 10 10

20 20 20

30 30 30

Kernel

1 0

0 -1

The kernel multiplies overlapping values, sums them, and produces one output value.

It then slides across the image, creating a new **feature map**.

14. Feature Map

After convolution, the output is called a **feature map**.

Image



Convolution



Feature Map

The feature map highlights important information such as:

- Vertical edges
 - Horizontal edges
 - Corners
 - Curves
 - Textures
-

15. Stride

Stride determines how many pixels the filter moves each step.

Example

Stride = 1

□□□□□



Moves

1 pixel

Stride = 2

□□□□□



Moves

2 pixels

Larger stride:

- Faster computation
 - Smaller output feature map
-

16. Padding

Padding adds extra pixels (usually zeros) around the image.

Without padding:

Input

5×5

↓

Output

3×3

With padding:

Input

5×5

↓

Add Border

↓

Output

5×5

Padding helps preserve edge information.

17. Why Multiple Filters?

One filter detects only one type of feature.

Example:

- Filter 1 → Vertical edges
- Filter 2 → Horizontal edges
- Filter 3 → Corners
- Filter 4 → Curves
- Filter 5 → Textures

Modern CNNs learn hundreds of filters automatically.

18. CNN Feature Learning

As layers go deeper, the learned features become more complex:

Layer 1



Edges



Layer 2



Corners



Layer 3



Textures



Layer 4



Eyes



Layer 5



Face



Output

Person

Key Takeaways

- A digital image is a matrix of pixel values.
- RGB images have **3 channels** (Red, Green, Blue).
- Images are represented as tensors (e.g., $224 \times 224 \times 3$).
- Flattening images for an ANN leads to too many parameters.
- CNNs use **convolution** to extract features efficiently.
- Kernels slide across the image to produce **feature maps**.
- **Stride** controls movement of the kernel.
- **Padding** preserves image size and edge information.
- CNNs automatically learn filters that detect increasingly complex features as the network gets deeper.

These concepts form the foundation for building image classification models starting from the next practical sessions.